Sungchul Lee

Ph.D. Student, HPCP Lab, Department of Computer Science, Yonsei University, Seoul, South Korea
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Education

Yonsei University Mar. 2024 – Present
Ph.D. Student in Computer Science and Engineering *Seoul, South Korea*

- Advisor: Prof. Youngsok Kim, HPCP Lab

Yonsei University Mar. 2018 – Feb. 2024
B.S. in Electrical and Electronic Engineering (GPA: 3.98/4.5) *Seoul, South Korea*

Publications

[5] **GJP: Fast and Efficient Join-Project Query Processing on GPUs Using Density-Aware Matrix Multiplication**
Sungchul Lee, Suhyun Lee, Hanna Cha, Chaemin Lim, and Youngsok Kim
*2027 ACM International Conference on Management of Data (**SIGMOD**)*, June 2027 (to appear)

[4] **GISim: Representative-Warp-Free GPU Interval Simulation**
Hanna Cha, **Sungchul Lee**, and Youngsok Kim
*33rd IEEE International Symposium on High-Performance Computer Architecture (**HPCA**)*, Mar. 2027 (to appear)

[3] **GCStack+GCScaler: Fast and Accurate GPU Performance Analyses Using Fine-Grained Stall Cycle Accounting and Interval Analysis**
Hanna Cha, **Sungchul Lee**, Jounghoo Lee, Yeonan Ha, Joonsung Kim, and Youngsok Kim
*52nd IEEE/ACM International Symposium on Computer Architecture (**ISCA**)*, June 2025

[2] **GCStack: A GPU Cycle Accounting Mechanism for Providing Accurate Insight Into GPU Performance**
Hanna Cha, **Sungchul Lee**, Yeonan Ha, Hanhwi Jang, Joonsung Kim, and Youngsok Kim
*IEEE Computer Architecture Letters (**CAL**)*, Oct. 2024

[1] **MPC-Wrapper: Fully Harnessing the Potential of Samsung Aquabolt-XL HBM2-PIM on FPGAs**
Jinwoo Choi[1], Yeonan Ha[1], Hanna Cha, Seil Lee, **Sungchul Lee**, Jounghoo Lee, Shin-haeng Kang, Bongjun Kim, Hanwoong Jung, Hanjun Kim, and Youngsok Kim
*32nd IEEE International Symposium on Field-Programmable Custom Computing Machines (**FCCM**)*, May 2024

Research Projects

DPU-Centric Datacenter Architecture for Next-Generation AI Devices Mar. 2024 – Present
Institute of Information & communications Technology Planning & evaluation (IITP)

Designing a PIM-Based NPU Architecture and FPGA Prototyping Mar. 2023 – Feb. 2024
Samsung Advanced Institute of Technology (SAIT)

Teaching Experience

Architecture of Computers – Teaching Assistant, Yonsei University Spring 2024

Discrete Mathematics – Teaching Assistant, Yonsei University Fall 2024